SMALL HODGE EIGENVALUES ON NODAL DEGENERATIONS OF THREEFOLDS

BERND JOHANNES WUEBBEN

Abstract. We determine the small positive Hodge eigenvalues on two-forms in a proper holomorphic family of compact threefolds over a disc, with smooth Kähler total space, smooth fibres away from the origin, and a reduced central fibre whose only singularities are ordinary double points. The fibre metrics are induced from the total space. Assume that a small resolution Y of the central fibre has vanishing first Betti number and second real cohomology spanned by first Chern classes of holomorphic line bundles. The number of small eigenvalues is the drop in the second Betti number from Y to a smooth fibre. Each is asymptotic to $2\pi\mu/\log(1/|t|)$, where t is the family parameter. The numbers μ are the relative eigenvalues of two quadratic forms on $H^2(Y;\mathbb{R})$ modulo the classes that extend to the smooth fibres: the sum of squared degrees on the exceptional curves, and the minimum squared L^2 norm of a harmonic representative on the regular part of the central fibre. We express the latter form through divisor intersections and the Kähler class. The remaining positive spectra in degrees one and two have a uniform gap, and the small eigenforms are coexact, primitive and of type $(1,1)$. The proof combines the norm calculation for a harmonic two-form of unit period on the node link, a sharp leading energy bound on the affine quadric smoothing, and the compactness theorem of Anné and Takahashi.

1. Introduction

1.1. **The result.** Let $\mathcal{X}$ be a connected complex manifold of dimension four with a Kähler form ω, let $\mathbb{D} \subset \mathbb{C}$ be a disc centred at the origin, and let

$$\pi : \mathcal{X} \longrightarrow \mathbb{D}$$

be a proper surjective holomorphic map with connected fibres $X_t = \pi^{-1}(t)$, which is a submersion over $\mathbb{D} \setminus \{0\}$. We assume that the fibre over the origin is reduced and that X_0 has exactly $N \geq 1$ singular points $p_1, \ldots, p_N$, each an ordinary double point. For $t \neq 0$ let g_t be the Kähler metric induced by ω on the smooth threefold X_t, and let

$$0 < \lambda^+_{2,1}(t) \leq \lambda^+_{2,2}(t) \leq \cdots$$

be the positive eigenvalues of its Hodge Laplacian $dd^* + d^*d$ on real two-forms, with multiplicity.

Let $q : Y \to X_0$ be a small resolution: Y is a complex manifold, q is an isomorphism over $X_0 \setminus \{p_1, \ldots, p_N\}$, and the $C_i = q^{-1}(p_i)$ are smooth rational curves. Small resolutions exist as complex manifolds, since each node has two local ones; Y need not be Kähler, and the hypotheses below do not require it. We assume

(1) $b_1(Y) = 0$, $\qquad H^2(Y;\mathbb{R})$ is spanned by first Chern classes of holomorphic line bundles.

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Both conditions hold, for instance, when Y is projective with $H^1(Y,\mathcal{O}_Y) = H^2(Y,\mathcal{O}_Y) = 0$. Put $V = H^2(Y;\mathbb{R})$ and $H = [q^*\omega] \in V$, and define

$$\operatorname{per} : V \longrightarrow \mathbb{R}^N, \qquad \operatorname{per}(D) = (D \cdot C_i)_i, \qquad K = \ker \operatorname{per},$$

$$G(D,E) = -\int_Y D\,E\,H + \frac{3\,(\int_Y D\,H^2)(\int_Y E\,H^2)}{2\int_Y H^3}. \tag{2}$$

The vector $\operatorname{per}(D)$ records the degrees of D on the exceptional curves. The kernel K is the image of $H^2(X_t;\mathbb{R})$ under restriction to the complement of small neighbourhoods of the nodes, which is common to all fibres, and it contains H (Lemma 3.1). We show that G is positive definite; it is the L^2 inner product of the harmonic representatives on the regular part of X_0, and on $\{\int_Y DH^2 = 0\}$ it agrees with the Hodge–Riemann form $-\int_Y D\,E\,H$ of H. These classes are G-orthogonal to H, and $G(H,H) = \frac{1}{2}\int_Y H^3$. Let $\mathcal{M}$ be the form induced by G on V/K, that is, the restriction of G to the G-orthogonal complement of K; equivalently

$$\mathcal{M}([D],[D]) = \min_{k\in K} G(D+k, D+k).$$

Let $\mathcal{E}([D],[E]) = \operatorname{per}(D)\cdot\operatorname{per}(E)$, and let

$$0 < \mu_1 \le \cdots \le \mu_m, \qquad m = \dim V/K = b_2(Y) - b_2(X_t),$$

be the eigenvalues of $\mathcal{E}$ relative to $\mathcal{M}$, that is, the roots of $\det(\mathcal{E} - \mu\mathcal{M}) = 0$. They do not depend on the choice of small resolution (Remark 6.3).

Theorem 1.1. *Under the preceding hypotheses,*

$$\lim_{t\to 0} \log(1/|t|)\,\lambda^+_{2,j}(t) = 2\pi\mu_j \qquad (1 \le j \le m). \tag{3}$$

There is $\gamma > 0$ such that, for all sufficiently small $t \neq 0$, the positive eigenvalues of the Hodge Laplacian on one-forms and the eigenvalues $\lambda^+_{2,j}(t)$ with $j > m$ are at least γ. For such t the eigenforms for $\lambda^+_{2,1}(t),\dots,\lambda^+_{2,m}(t)$ are coexact, primitive and of type $(1,1)$.

Thus the number of small eigenvalues is the number of independent classes in $H^2(Y)$ that do not extend to the smooth fibres, and the limits are computed from the intersection form of Y, the class of the metric and the degrees of these classes on the exceptional curves. When $m = 1$ and $K = \mathbb{R}H$,

$$\lambda^+_{2,1}(t) = \frac{2\pi\sum_i (D\cdot C_i)^2}{(\int_Y DH^2)^2/\int_Y H^3 - \int_Y D^2H}\;\frac{1+o(1)}{\log(1/|t|)} \tag{4}$$

for any class D independent of H. For the determinantal hypersurface $\{l_1f_2 - l_2f_1 = 0\} \subset \mathbb{P}^4$ of degree $d \ge 2$, with $(d-1)^2$ nodes, smoothed in a pencil and given the Fubini–Study metric with lines of area one, the coefficient is $2\pi d$ (Example 6.1).

1.2. **Relation to earlier work.** Anné and Takahashi [AT15] study the Hodge–de Rham operator on a manifold in which a compact piece with conical boundary is scaled to zero. They determine the multiplicity of zero in the limit spectrum and obtain convergence of the remaining spectrum; in degrees related to the cohomology of the link they construct test forms of energy $O(1/|\log\epsilon|)$ in the scaling parameter ϵ, which bounds certain eigenvalues above. We use their compactness results [AT15, Corollary 17 and Lemma 18]. The leading constants of such eigenvalues are not determined there.

In [W26] the limits (3) are proved, along rays $t \to 0$, for the Ricci-flat Kähler metrics in a polarized Calabi–Yau conifold transition, with the same forms $\mathcal{E}$ and $\mathcal{M}$. That proof uses the description of the Ricci-flat metrics near the singular fibre due to Hein and Sun [HS17] and

Benabida [Ben26]. For the induced metrics considered here the geometry near the nodes is explicit (Lemma 2.1), no condition on the canonical bundle is needed, and the limit is taken over all $t \to 0$, not only along rays. The tangent cone at a node, its link, the local smoothing of the cone, and the constants attached to them differ from the Ricci-flat case, while the limits agree whenever both theorems apply (Remark 6.2); the limits depend on the Kähler metric only through its class, because the squared norm of the harmonic form on the link and the logarithm of the scale of the local smoothing change by the same factor (Remark 6.2). The statement about types is proved differently here, without a parallel holomorphic volume form (Proposition 5.2).

The spectral argument of Section 4 adapts that of [W26, Sections 4–6] to the cone $\{\sum_j \sigma_j w_j^2 = 0\} \subset \mathbb{C}^4$ of Section 2; we give it in full, so that no result of [W26] is used. New here are the models of Section 2, the construction of the harmonic representatives without a Ricci-flat background (Proposition 3.2), the passage to arbitrary $t \to 0$ (Lemma 2.5), and the proof of the statement about types.

For functions, small eigenvalues of metrics induced on degenerating families are studied by Dai and Yoshikawa [DY26] for curves and by Cao [Cao26] in all dimensions; they have order $1/\log(1/|t|)$, and for a reduced central fibre their number is the number of its components minus one [DY26, Theorem 0.2], [Cao26, Theorem 0.4]. Dai and Yoshikawa raise the question of small eigenvalues for forms of higher degree: for $(n,0)$-forms with induced metrics and for forms of type (p,q) with Kähler–Einstein metrics of negative scalar curvature [DY26, Conjecture 9.5 and Problem 9.7]. Theorem 1.1 is a result of this kind in a case not covered by those formulations, two-forms of type $(1,1)$ with induced metrics on threefolds acquiring ordinary double points; here the central fibre is irreducible and there are no small eigenvalues on functions.

1.3. **Method.** In suitable holomorphic coordinates at a node the map π is a diagonal quadratic form $\sum_j \sigma_j w_j^2$ and ω is the flat Kähler form up to terms vanishing at the node. For the flat form, the fibres of π near the node are exact rescalings, by $|t|^{1/2}$, of the affine quadric $\{\sum_j \sigma_j w_j^2 = 1\}$, and the central fibre is the cone over a circle bundle L_σ over a quadric surface. The harmonic two-form on L_σ representing a generator of $H^2(L_\sigma;\mathbb{Z})$ has squared L^2 norm π, independently of σ (Lemma 2.2).

A class $D \in V$ has a closed and coclosed representative on the regular part of X_0, asymptotic to $(D \cdot C_i)$ times that form at the ith node (Proposition 3.2). It extends to a closed form on X_t only if all these periods vanish. Otherwise, cutting it off between the scale $|t|^{1/2}$ of the rescaled affine quadric and a fixed scale costs energy, that is $\|d\psi\|^2 + \|d^*\psi\|^2$, equal to $\pi \sum_i (D \cdot C_i)^2 / \log(1/|t|^{1/2})$ to leading order, and a lower bound for the energy on the affine quadric shows that no form with the same periods has less (Lemma 4.1). The limits (3) follow from this on the model manifolds of Definition 2.4, which are exactly conical near the nodes (Theorem 4.6), using the compactness theorem of [AT15], and are transferred to the metrics g_t by Dodziuk's comparison of Hodge eigenvalues for uniformly equivalent metrics [Dod82], which requires only that g_t be close to the models in C^0 (Lemma 2.5). The statement about types follows because the projection onto forms of type $(2,0)+(0,2)$ commutes with the Laplacian of a Kähler metric, while the limits of the small eigenforms are of type $(1,1)$.

1.4. **Organization.** Section 2 contains the local geometry and the models, Section 3 the harmonic representatives and the form G, Section 4 the spectral analysis of the models, and Section 5 the proof of Theorem 1.1. Section 6 contains the example and remarks.

2. Geometry near a node

Throughout, $d^c = \frac{i}{2}(\bar\partial - \partial)$, so that $dd^c = i\partial\bar\partial$, and for a Kähler form ω and a real function f the contraction $\Lambda_\omega dd^c f$ equals $-\frac{1}{2}\Delta f$, where $\Delta = d^*d$. The Hodge Laplacian on forms is $\Delta = dd^* + d^*d$. The letter C denotes constants independent of the family parameter t and of the auxiliary radii R and A introduced below; $c(R, A)$ denotes constants that may depend on them.

2.1. **Normal form.** Let p be a singular point of X_0. Since $\mathcal{X}$ is smooth and the fibre $\pi^{-1}(0)$ is reduced, π is a local equation of X_0 at p; as X_0 has an ordinary double point there, π has a nondegenerate critical point at p.

Lemma 2.1. *There are holomorphic coordinates $w = (w_1, \dots, w_4)$ on a neighbourhood U_p of p in $\mathcal{X}$, centred at p, and numbers $\sigma_1, \dots, \sigma_4 > 0$, such that*

$$\pi = Q_\sigma(w) := \sum_{j=1}^4 \sigma_j w_j^2, \qquad \omega = \omega_{\mathrm{fl}} + \omega', \quad \omega_{\mathrm{fl}} = \tfrac{i}{2}\sum_j dw_j \wedge d\bar w_j, \tag{5}$$

where ω' is a smooth $(1,1)$-form on U_p vanishing at p.

Proof. By the holomorphic Morse lemma there are coordinates z with $\pi = \sum_j z_j^2$. Let $\omega_p = \frac{i}{2}\sum h_{j\bar k} dz_j \wedge d\bar z_k$ at p, with $(h_{j\bar k})$ positive Hermitian, and choose an invertible matrix T such that the linear change $z = Tw'$ brings ω_p to $\frac{i}{2}\sum dw'_j \wedge d\bar w'_j$. In these coordinates $\pi = (w')^{\mathrm t} S w'$ with $S = T^{\mathrm t}T$ complex symmetric and invertible. By the Autonne–Takagi factorization [HJ13, Section 4.4] there is a unitary U with $S = U^{\mathrm t}\,\mathrm{diag}(\sigma_j)U$ and $\sigma_j \ge 0$; invertibility gives $\sigma_j > 0$. Put $w = Uw'$. □

We fix such coordinates at every node p_i, $1 \le i \le N$, with coefficients $\sigma^{(i)}$, and write $r = |w|$ in each chart. Thus, in the chart,

$$X_t \cap U_p = \{Q_\sigma = t\} \cap U_p \qquad (t \in \mathbb{D}), \tag{6}$$

and $X_0 \cap U_p$ is an open subset of the cone $C_\sigma = \{Q_\sigma = 0\} \subset \mathbb{C}^4$.

2.2. **The cone and its link.** Let g_C be the metric induced on $C_\sigma \setminus \{0\}$ by the flat metric of $\mathbb{C}^4$. It is a Kähler cone metric $g_C = dr^2 + r^2 h$ with Kähler form $\omega_C = \omega_{\mathrm{fl}}|_{C_\sigma}$ and link $L_\sigma = C_\sigma \cap S^7$ with the induced metric h. The circle $\{e^{i\phi}\}$ acts freely and isometrically on L_σ with orbits of length 2π, and the quotient is the smooth quadric surface $Z_\sigma = \{Q_\sigma = 0\} \subset \mathbb{P}^3$. Let $\varpi : L_\sigma \to Z_\sigma$ be the projection. It is a Riemannian submersion onto Z_σ with the Kähler form ω_Z given by $\varpi^*\omega_Z = \frac{i}{2}\partial\bar\partial \log|w|^2$, and the one-form $\xi = d^c \log|w|^2$ restricts to $d\phi$ on the orbits and satisfies $d\xi = 2\varpi^*\omega_Z$.

The surface Z_σ is isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$. Let $F_1, F_2 \in H^2(Z_\sigma; \mathbb{Z})$ be the classes of the fibres of its two rulings, so that $F_1^2 = F_2^2 = 0$, $F_1F_2 = 1$, and the hyperplane class is $F_1 + F_2$. The link is the unit circle bundle of $\mathcal{O}(-1)|_{Z_\sigma}$, and the Gysin sequence gives

$$H^1(L_\sigma; \mathbb{Z}) = 0, \qquad H^2(L_\sigma; \mathbb{Z}) = H^2(Z_\sigma; \mathbb{Z})/\mathbb{Z}(F_1 + F_2) \cong \mathbb{Z}, \qquad H^3(L_\sigma; \mathbb{Z}) \cong \mathbb{Z}. \tag{7}$$

The group $H^2(L_\sigma; \mathbb{Z})$ is generated by $\varpi^*F_1 = -\varpi^*F_2$. Which of the two generators is used at a node is fixed in Section 3 by the small resolution.

Lemma 2.2. *There is a unique harmonic two-form ϑ on (L_σ, h) in the class ϖ^*F_1. It satisfies*

$$\|\vartheta\|^2_{L^2(L_\sigma, h)} = \pi \tag{8}$$

for every σ. Its pull-back to $C_\sigma \setminus \{0\}$ under the radial projection, again denoted ϑ, is a closed and coclosed real primitive $(1,1)$-form for (g_C, ω_C), with $|\nabla_C^j \vartheta|_{g_C} = O(r^{-2-j})$ for all $j \geq 0$ and vanishing contraction with ∂_r. The form $*_h\vartheta$ is closed and generates $H^3(L_\sigma;\mathbb{R})$.

Proof. Let ϑ_Z be the harmonic representative on (Z_σ, ω_Z) of the class $e = \frac{1}{2}(F_1 - F_2)$. Since $H^{2,0}(Z_\sigma) = 0$ it is a real $(1,1)$-form, and it is primitive because $e \cdot (F_1 + F_2) = 0$ and ω_Z is a multiple of the hyperplane class. On a Kähler surface a primitive real $(1,1)$-form is anti-self-dual, so

$$\int_{Z_\sigma} |\vartheta_Z|^2 \, dV = -\int_{Z_\sigma} \vartheta_Z \wedge \vartheta_Z = -e^2 = \tfrac{1}{2}.$$

Put $\vartheta = \varpi^*\vartheta_Z$. It is closed, and its class is $\varpi^* e = \varpi^* F_1$ because $\varpi^*(F_1 + F_2) = 0$. For the Riemannian submersion ϖ with unit vertical one-form ξ, the Hodge star of the pull-back of a two-form β is $\varpi^*(*_Z\beta) \wedge \xi$ for the orientation $\varpi^* dV_Z \wedge \xi$. Hence $*_h\vartheta = -\vartheta \wedge \xi$, and $d(\xi \wedge \vartheta) = 2\varpi^*(\omega_Z \wedge \vartheta_Z) = 0$ by primitivity. Thus ϑ is harmonic; uniqueness follows from $b_2(L_\sigma) = 1$. Since the fibres of ϖ have length 2π,

$$\|\vartheta\|^2_{L^2(L_\sigma)} = 2\pi \int_{Z_\sigma} |\vartheta_Z|^2 \, dV = \pi.$$

On the cone, ϑ is the pull-back of ϑ_Z under the holomorphic map $C_\sigma \setminus \{0\} \to Z_\sigma$, hence closed, real, of type $(1,1)$ and homogeneous of degree zero, which gives the stated bounds and $\iota_{\partial_r}\vartheta = 0$. With $\omega_C = \frac{1}{2} d(r^2\xi) = r\, dr \wedge \xi + r^2 \varpi^*\omega_Z$ one finds $\vartheta \wedge \omega_C^2 = 0$, because $\vartheta_Z \wedge \omega_Z = 0$ and $\vartheta \wedge \varpi^*\omega_Z^2$ is the pull-back of a six-form from a surface. So ϑ is primitive, and then $*\vartheta = -\vartheta \wedge \omega_C$ is closed. Finally $\int_{L_\sigma} \vartheta \wedge *_h\vartheta = \pi \neq 0$, so the closed form $*_h\vartheta$ is not exact. □

2.3. **The smoothing cap.** Let $V_\sigma = \{Q_\sigma = 1\} \subset \mathbb{C}^4$ with the metric g_V induced by the flat metric. It is diffeomorphic to T^*S^3; in particular

$$H^1(V_\sigma) = H^2(V_\sigma) = 0, \qquad H^3(V_\sigma;\mathbb{R}) \xrightarrow{\sim} H^3(L_\sigma;\mathbb{R}), \tag{9}$$

the second map being restriction to a large sphere. For $t = \epsilon^2 u^2$ with $\epsilon > 0$ and $|u| = 1$,

$$\{Q_\sigma = t\} = \epsilon u \cdot V_\sigma, \tag{10}$$

and multiplication by ϵu multiplies the flat metric by ϵ^2. Throughout we write

$$\epsilon = |t|^{1/2}, \qquad u^2 = t/|t|. \tag{11}$$

Lemma 2.3. *Put $n(w) = \overline{\partial Q_\sigma(w)}/|\partial Q_\sigma(w)|^2$, where $\partial Q_\sigma = (2\sigma_j w_j)_j$. There are $A_0 > 0$ and a smooth complex-valued function τ on $C_\sigma \cap \{r > A_0\}$ with $\tau = 1 + O(r^{-2})$ such that*

$$\Psi(w) = w + \tau(w)\, n(w)$$

is a diffeomorphism from $C_\sigma \cap \{r > A_0\}$ onto the complement of a compact subset of V_σ, and

$$|\nabla_C^j(\Psi^* g_V - g_C)|_{g_C} \leq C_j r^{-2-j} \qquad (j \geq 0). \tag{12}$$

Proof. Since Q_σ is quadratic, $Q_\sigma(w+\tau n) = Q_\sigma(w) + \tau + \tau^2 Q_\sigma(n)$, because $dQ_\sigma(w)\cdot n(w) = 1$. The vector field n is homogeneous of degree -1 under positive dilations and $|n| \leq Cr^{-1}$, so $Q_\sigma(n) = O(r^{-2})$ with corresponding bounds on derivatives, and the equation $\tau + \tau^2 Q_\sigma(n) = 1$ has a unique smooth solution near 1 for large r, with $\tau - 1 = O(r^{-2})$. The displacement τn and its covariant derivatives of order j are $O(r^{-1-j})$, so Ψ differs from the inclusion of the cone by a map whose first derivatives are $O(r^{-2})$; this gives (12). The same estimate shows that Ψ is an injective immersion for large r, and its image contains every point $v \in V_\sigma$ of

large norm: solving $Q_\sigma(v-\tau' n(v))=0$ in the same way gives the cone point over v, up to a correction that is removed by the inverse function theorem. □

2.4. **Conic surgery models.** Let $M_0 = X_0 \setminus \{p_1,\dots,p_N\}$ with the Kähler metric g_0 induced by ω. By (5), on the end of M_0 at p_i,

$$|\nabla_C^j (g_0 - g_C)|_{g_C} \le C_j r^{1-j} \qquad (j \ge 0), \tag{13}$$

and likewise for the Kähler forms. Fix $R_0 > 0$ such that the balls $\{r \le 4R_0\}$ at the nodes lie in the charts and are pairwise disjoint.

Fix $0 < R \le R_0$ and $A \ge A_0$. Let g_0^R be a smooth metric on M_0 equal to g_0 where $r \ge 2R$ and to g_C where $r \le R$ at every node, obtained by interpolation with a fixed cutoff in r/R. On each $V_{\sigma^{(i)}}$ let g_V^A be a smooth metric equal to g_V on the compact set bounded by $\Psi(\{r=A\})$ and to $(\Psi^{-1})^* g_C$ outside $\Psi(\{r=2A\})$, interpolated in the same way. By (13) and (12),

$$|g_0^R - g_0|_{g_0} \le CR, \qquad |g_V^A - g_V|_{g_V} \le CA^{-2}. \tag{14}$$

These metrics need not be Kähler.

For $T > A_0$ let V_σ^T be the open subset of V_σ with compact closure bounded by $\Psi(\{r=T\})$.

Definition 2.4. Let $8A\epsilon < R$ and $|u|=1$. The *conic surgery model* $(\overline{X}_{\epsilon,u}, \bar g_{\epsilon,u}^{R,A})$ is the closed Riemannian manifold obtained from the central piece $(M_0 \cap \{r > 2A\epsilon\}, g_0^R)$ and, for each node, the cap core $(V_\sigma^{4A}, \epsilon^2 g_V^A)$ with $\sigma = \sigma^{(i)}$, by identifying each point w of the central piece with $2A\epsilon < |w| < 4A\epsilon$ in the chart at the node with the point $\Psi(w/\epsilon u)$ of the cap core.

The identification is an isometry between open collars on which both metrics are exactly conical, so the model is a smooth closed Riemannian manifold. At each node the part of the model inside a section $\{r=T\}$, $4A\epsilon < T < R$, is the union of the cap core and the conical region $\{2A\epsilon < r \le T\}$; the map $v \mapsto \epsilon u\, \Psi^{-1}(v)$ on the conical part of the cap identifies it isometrically with the truncated cap $(\overline{V_\sigma^{T/\epsilon}}, \epsilon^2 g_V^A)$.

Lemma 2.5. *Let $t = \epsilon^2 u^2$. For $|t|$ small there are diffeomorphisms $\Phi_t : \overline{X}_{\epsilon,u} \to X_t$ with*

$$\limsup_{t\to 0} \sup_{\overline{X}_{\epsilon,u}} \left|\Phi_t^* g_t - \bar g_{\epsilon,u}^{R,A}\right|_{\bar g_{\epsilon,u}^{R,A}} \le C\,(R + A^{-2}). \tag{15}$$

Moreover, for unit complex numbers u, u' there is a diffeomorphism $\overline{X}_{\epsilon,u} \to \overline{X}_{\epsilon,u'}$ under which the two model metrics differ by at most $C|u-u'|$ relative to either of them; here and below, two metrics $g, \bar g$ differ by at most δ relative to $\bar g$ if $(1-\delta)\bar g \le g \le (1+\delta)\bar g$.

Proof. On the cap core at a node define $\Phi_t(v) = \epsilon u v$, which lies in $\{Q_\sigma = t\} = X_t \cap U_p$ by (10). On the conical region $\{2A\epsilon < |w| < R\}$ of the central piece put

$$\Psi_t(w) = \epsilon u\, \Psi(w/\epsilon u) = w + t\,\tau_t(w)\, n(w), \qquad \tau_t + t\tau_t^2 Q_\sigma(n(w)) = 1,$$

which is compatible with the gluing of the model, takes values in $\{Q_\sigma = t\}$, and depends smoothly on (t,w) for $|w| \ge R/2$ and small t, with Ψ_0 the identity. The map $(t,w) \mapsto \Psi_t(w)$ is an open embedding of $\mathbb{D} \times \{R/2 < |w| < R\}$ into $\mathcal{X}$ over $\mathbb{D}$, for a smaller disc. Push the two real coordinate vector fields of $\mathbb{D}$ forward by it, and extend them, by a partition of unity, to lifts of these vector fields on a neighbourhood of $M_0 \cap \{r \ge R/2\}$ in $\mathcal{X}$, where π is a submersion. Flowing along the two lifts successively extends Ψ_t to embeddings of $M_0 \cap \{r > R/2\}$ into X_t, smooth in t and equal to the identity at $t=0$, whose image is the complement of the image of $\{|w| \le R/2\}$ and of the cap cores. This defines Φ_t.

On the exterior $\{r \geq R\}$ the metrics $\Phi_t^* g_t$ converge uniformly to g_0, and g_0^R differs from g_0 by at most CR by (14). On the cap core and the conical region, identified with a truncated cap as after Definition 2.4, the map Φ_t is $v \mapsto \epsilon u v$; the form $\Phi_t^* \omega_{\mathrm{fl}}$ restricts to ϵ^2 times the Kähler form of g_V, and by (5) the form ω' is bounded by $C|w| \leq CR$ relative to ω_{fl} on the region $|w| \leq R$ containing the image. Hence $\Phi_t^* g_t$ differs from $\epsilon^2 g_V$ by at most CR there, and g_V^A differs from g_V by at most CA^{-2}. This proves (15).

For the last assertion write $u' = e^{i\delta} u$ and let χ be a fixed cutoff, equal to one for $r \leq R/2$ and to zero for $r \geq R$. The map of the central piece equal to $w \mapsto e^{i\delta\chi(|w|)} w$ in the charts and to the identity elsewhere preserves the cone C_σ, and together with the identity on the cap cores it is compatible with the two gluing maps, which involve only the region $\{r < 4A\epsilon\}$, where $\chi = 1$. Multiplication by a unit scalar is an isometry of g_C, and the radial dependence of the angle changes the metric g_0^R by at most $C|\delta|$. □

3. Topology and central harmonic forms

3.1. **The restriction map.** Let $q : Y \to X_0$ be the small resolution, with exceptional curves $C_i = q^{-1}(p_i) \cong \mathbb{P}^1$, and identify M_0 with $Y \setminus \bigcup_i C_i$. Removing submanifolds of real codimension four does not change cohomology in degrees one and two, so by hypothesis

$$H^1(M_0; \mathbb{R}) = 0, \qquad H^2(M_0; \mathbb{R}) = H^2(Y; \mathbb{R}) = V. \tag{16}$$

A neighbourhood of C_i in Y retracts onto C_i, and its boundary is the link $L_i = L_{\sigma^{(i)}}$. The restrictions $H^2(N) \to H^2(C_i)$ and $H^2(N) \to H^2(L_i)$ of such a neighbourhood N are isomorphisms, and we let ϑ_i be the form of Lemma 2.2 for the generator of $H^2(L_i; \mathbb{Z})$ that corresponds to the positive generator of $H^2(C_i; \mathbb{Z})$. With this convention the restriction of a class $D \in V$ to L_i is $(D \cdot C_i)[\vartheta_i]$. Define

$$\mathrm{per} : V \longrightarrow \mathbb{R}^N, \qquad \mathrm{per}(D) = (D \cdot C_i)_i, \qquad K = \ker \mathrm{per}, \qquad \mathcal{Q} = V/K. \tag{17}$$

Lemma 3.1. *Let W be the complement in M_0 of small open cone neighbourhoods of the nodes, identified with a subset of every X_t by Lemma 2.5. For $t \neq 0$ restriction to W identifies $H^2(X_t; \mathbb{R})$ with K, and $b_1(X_t) = 0$. In particular $b_2(X_t) = s := \dim K$, and $m := \dim \mathcal{Q} = b_2(Y) - b_2(X_t)$.*

Proof. By (6) and Lemma 2.5, X_t is obtained from W by attaching copies of a compact truncation of V_σ along the links. Apply the Mayer–Vietoris sequence, using (7), (9) and (16): the maps in degree zero are surjective and the terms in degree one vanish, so $b_1(X_t) = 0$; in degree two $H^2(X_t)$ is the kernel of the restriction $H^2(W) = V \to \bigoplus_i H^2(L_i)$, which is K. This is the classical description of the cohomology of a conifold transition [Ros06, Theorem 3.2], [LLW18, Section 1]. □

The class $H = [q^* \omega]$ lies in K, because $q^* \omega$ vanishes on the exceptional curves. Put

$$B(D, E) = \int_Y D\, E\, H, \qquad a(D) = \int_Y D\, H^2, \qquad v = \int_Y H^3. \tag{18}$$

Since $q^* \omega$ is a smooth closed form on Y in the class H and the curves C_i have measure zero,

$$\int_{M_0} \omega^3 = v, \qquad \mathrm{Vol}(M_0, g_0) = \frac{v}{6}. \tag{19}$$

3.2. **Harmonic representatives.**

Proposition 3.2. *For every $D \in V$ there is a smooth real closed $(1,1)$-form η_D on M_0 in the class D, depending linearly on D, such that:*

(1) *η_D is coclosed for g_0, square integrable, and $\Lambda_\omega \eta_D = 3a(D)/v$;*
(2) *for some $\nu \in (0,1)$ and all $j \geq 0$, on the end at p_i,*

$$|\nabla_C^j(\eta_D - \mathrm{per}_i(D)\vartheta_i)|_{g_C} \leq C_j r^{\nu-2-j} \, \|D\|; \tag{20}$$

(3) *for $D, E \in V$,*

$$\langle \eta_D, \eta_E \rangle_{L^2(M_0,g_0)} = G(D,E) = -B(D,E) + \frac{3a(D)a(E)}{2v}, \tag{21}$$

with G the form of (2). In particular G is positive definite on V. Here $\|D\|$ is any fixed norm on V, and $\mathrm{per}_i(D) = D \cdot C_i$.

Proof. By hypothesis V is spanned by first Chern classes of holomorphic line bundles on Y; it suffices to construct η_D for such a class and to extend linearly. Let $\mathcal{L}$ be a line bundle with $c_1(\mathcal{L}) = D$ and let θ_b be the normalized curvature form of a smooth Hermitian metric on $\mathcal{L}$.

A model form near the nodes. Let $N_i = q^{-1}(\{r < R_0\})$ be the neighbourhood of C_i in Y. An ordinary double point has exactly two small resolutions, each with exceptional curve of normal bundle $\mathcal{O}(-1)^{\oplus 2}$, so N_i is isomorphic to a neighbourhood of the zero section in $\mathcal{O}(-1)^{\oplus 2} \to \mathbb{P}^1$. Since q maps N_i properly onto a Stein space with $R^1 q_* \mathcal{O} = 0$, we have $H^1(N_i, \mathcal{O}) = 0$, and a holomorphic line bundle on N_i is determined up to isomorphism by its degree on C_i: it is a power of the pull-back $\mathcal{R}_i$ of $\mathcal{O}_{\mathbb{P}^1}(1)$. Thus $\mathcal{L}|_{N_i} \cong \mathcal{R}_i^{b_i}$ with $b_i = D \cdot C_i$. On $N_i \setminus C_i$, identified with a punctured neighbourhood of the vertex of C_σ, the bundle $\mathcal{R}_i$ is the pull-back under $C_\sigma \setminus \{0\} \to Z_\sigma$ of the line bundle of one ruling, whose class is the generator used to define ϑ_i. Let h_Z be a Hermitian metric on that bundle over Z_σ whose normalized curvature is the harmonic representative of its class, namely $\vartheta_Z + \frac{1}{2\pi}\omega_Z$ in the notation of the proof of Lemma 2.2. The pulled-back metric multiplied by $|w|$ has normalized curvature $\vartheta_i + \frac{1}{2\pi}\varpi^*\omega_Z - \frac{1}{2\pi}\frac{i}{2}\partial\bar\partial \log|w|^2 = \vartheta_i$. Hence $\mathcal{L}|_{N_i \setminus C_i}$ carries a Hermitian metric with normalized curvature $b_i\vartheta_i$. Its logarithmic ratio u_i to the smooth metric satisfies $u_i = O(1 + |\log r|)$ and $|\nabla_C^j u_i| = O(r^{-j})$ for $j \geq 1$, and θ_b has size $O(r^{-2})$ in the cone metric, with corresponding bounds for derivatives: indeed the two metrics on $\mathcal{R}_i$ over the punctured cone differ by the factor $|w|$ and by a function that is smooth on N_i or homogeneous of degree zero, and a function or form that is smooth on Y has cone derivatives of order j bounded by Cr^{-j}, respectively Cr^{-2-j}, because the dilations of the cone extend to N_i. Patching the u_i with cutoffs supported in the N_i and equal to one near the C_i we obtain a function u on M_0 and a real closed $(1,1)$-form

$$\theta = \theta_b + \tfrac{1}{2\pi} dd^c u,$$

where u is the logarithm of the ratio of the smooth metric to the patched one, which represents D on M_0 and equals $b_i\vartheta_i$ near p_i. Since ϑ_i is primitive for the cone and $\omega - \omega_C = O(r)$ by (13),

$$|\nabla_C^j \Lambda_\omega \theta| \leq C_j r^{-1-j} \qquad \text{near the nodes.} \tag{22}$$

Intersection numbers. Because θ_b and $q^*\omega$ are smooth closed forms on Y,

$$\int_{M_0} \theta_b \wedge \omega^2 = a(D), \qquad \int_{M_0} \theta_b^2 \wedge \omega = B(D,D).$$

Truncate the ends at $r=\rho$. Replacing θ_b by θ changes the first integral by the boundary term $\int_{r=\rho} d^c u\wedge\omega^2 = O(\rho^4)$, and the second by $\int_{r=\rho} d^c u\wedge(\theta+\theta_b)\wedge\omega = O(\rho^2)$, up to constant factors; the integrands are absolutely integrable, their worst size being $O(r^{-4})$ against the volume density $r^5 dr$. Hence

$$\int_{M_0}\theta\wedge\omega^2 = a(D), \qquad \int_{M_0}\theta^2\wedge\omega = B(D,D). \tag{23}$$

The scalar correction. The space (X_0, g_0) is a compact space with isolated conical singularities in the sense of [Joy04, Definition 2.1], by (13), after replacing r by the distance to the node, and M_0 is connected. The function $\varrho = 3a(D)/v - \Lambda_\omega\theta$ has integral zero, because $\Lambda_\omega\theta\, dV = \frac{1}{2}\theta\wedge\omega^2$ and (19), (23); and it is $O(r^{-1})$ with all derivatives by (22). The exponents β of homogeneous harmonic functions $r^\beta\phi$ on a six-dimensional cone are the roots of $\beta(\beta+4)=\lambda_L$ for eigenvalues $\lambda_L\geq 0$ of the link, so none lies in $(-4,0)$ and the positive ones are bounded below by some $\beta_1>0$. Fix $\nu\in(0,\min(1,\beta_1))$. By [Joy04, Proposition 2.18 and Theorem 2.9] there is a smooth function f on M_0 with

$$\Lambda_\omega dd^c f = \varrho, \qquad |\nabla_C^j(f-f_i)| \leq C_j r^{\nu-j} \ \text{ on the end at } p_i, \tag{24}$$

for constants f_i. Put $\eta_D = \theta + dd^c f$.

Properties. The form η_D is real, closed, of type $(1,1)$, represents D, has $\Lambda_\omega\eta_D = 3a(D)/v$, and satisfies (20) by (24). Write $\eta_D = \alpha + (a(D)/v)\,\omega$. Then α is closed and primitive, so $*\alpha = -\alpha\wedge\omega$ is closed, and $*\omega = \frac{1}{2}\omega^2$ is closed; thus η_D is coclosed. It is square integrable because $|\eta_D| = O(r^{-2})$ and the volume density is r^5dr. The boundary terms produced by replacing θ with η_D in (23) are $O(\rho^{\nu+4})$ and $O(\rho^{\nu+2})$, so $\int\eta_D\wedge\omega^2 = a(D)$ and $\int\eta_D^2\wedge\omega = B(D,D)$. Since α is primitive and pointwise orthogonal to ω, and $\|\omega\|^2 = 3\operatorname{Vol}(M_0) = v/2$,

$$\|\eta_D\|^2 = -\int_{M_0}\alpha^2\wedge\omega + \frac{a(D)^2}{v^2}\cdot\frac{v}{2} = -B(D,D) + \frac{a(D)^2}{v} + \frac{a(D)^2}{2v},$$

which is (21) on the diagonal; polarization gives the general case. Finally, $G(D,D)=0$ forces $\eta_D = 0$ and hence $D=0$. □

Lemma 3.3. *Let g be a smooth metric on M_0 uniformly equivalent to g_C near each node. A smooth square-integrable two-form on M_0 that is closed and coclosed for g and exact is zero.*

Proof. Let ψ be such a form. On an end write $\psi = a(r) + dr\wedge b(r)$ with a, b forms on the link, choose v_0 on the link with $d_L v_0 = a(r_0)$, and put $\beta(r) = v_0 + \int_{r_0}^r b(x)\,dx$. Then $d\beta = \psi$ on the end. With $B_0(x) = \|b(x)\|_{L^2(L)}$, square integrability and cone equivalence give $\int_0^{r_0} x^3 B_0(x)^2 dx < \infty$, and for $0<r<\delta<r_0$ the Cauchy–Schwarz inequality gives

$$r\|\beta(r)\|_{L^2(L)} \leq rC_\delta + 2^{-1/2}\Big(\int_0^\delta x^3 B_0(x)^2 dx\Big)^{1/2}.$$

Hence $r\|\beta(r)\|_{L^2(L)}\to 0$ as $r\to 0$. In particular β is square integrable, and for a cutoff χ_ρ vanishing below ρ, equal to one above 2ρ, with $|d\chi_\rho|\leq C/\rho$,

$$\|d\chi_\rho\wedge\beta\|^2 \leq C\rho^{-2}\int_\rho^{2\rho} r^3\|\beta(r)\|^2_{L^2(L)}dr \longrightarrow 0.$$

Let γ be a global smooth primitive of ψ. On each end $\gamma-\beta$ is closed, hence exact because $H^1(L)=0$; subtracting the differential of a cutoff of its potential we may assume $\gamma=\beta$ near

the nodes. Then $\chi_\rho\gamma$ is compactly supported and

$$\|\psi\|^2 = \lim_{\rho\to 0}\langle d(\chi_\rho\gamma),\psi\rangle = \lim_{\rho\to 0}\langle \chi_\rho\gamma, d^*\psi\rangle = 0.$$

□

Consequently, for g_0 and for each g_0^R, a smooth square-integrable closed and coclosed two-form on M_0 is determined by its class in V. For g_0 the space of such forms is $\{\eta_D : D \in V\}$.

3.3. **The quotient form.** Let $\mathcal{M}$ be the quotient of G on $\mathcal{Q} = V/K$,

$$\mathcal{M}([D],[D]) = \min_{k\in K} G(D+k, D+k), \tag{25}$$

and let $\mathcal{E}([D],[E]) = \mathrm{per}(D)\cdot\mathrm{per}(E)$, the Euclidean inner product in $\mathbb{R}^N$. Both are positive definite on the m-dimensional space $\mathcal{Q}$. Let

$$0 < \mu_1 \le \cdots \le \mu_m$$

be the eigenvalues of $\mathcal{E}$ relative to $\mathcal{M}$. Since $G(D,H) = a(D)/2$, the G-orthogonal complement of K is contained in $\{a(D)=0\}$, where $G = -B$. In a basis of K followed by lifts of a basis of $\mathcal{Q}$ the matrix of $\mathcal{M}$ is the Schur complement

$$\mathsf{M} = -B_{QQ} + B_{QK}B_{KK}^{-1}B_{KQ}; \tag{26}$$

here B_{KK} is invertible because it differs from the negative definite matrix $-G_{KK}$ by the rank-one term $\frac{3}{2v}aa^T$ and $a^TG_{KK}^{-1}a = 4G(H,H) = 2v$. When $m = 1$ and $K = \mathbb{R}H$, a class D spanning a complement gives

$$\mu_1 = \frac{\sum_i (D\cdot C_i)^2}{a(D)^2/v - B(D,D)}. \tag{27}$$

4. The spectrum of the conic surgery models

In this section R, A and the unit u are fixed, and we study the models $(\overline{X}_{\epsilon,u}, \bar g_{\epsilon,u}^{R,A})$ of Definition 2.4 as $\epsilon\to 0$. We abbreviate them $(\overline{X}_\epsilon, \bar g_\epsilon)$ and write $b = \dim V$, $s = \dim K$, $m = b - s$, $c = \pi$ for the constant of (8), and $\ell_\epsilon = \log(1/\epsilon)$. The arguments follow [W26, Sections 4–6], where the cone is the Ricci-flat conifold; we give them in full, since the cone, its link and the cap are different here and only the properties recorded in Section 2 are used.

4.1. **The energy of a period across a cap.** For a smooth two-form ψ on a region of a cone or of a cap that contains the section $\{r = T\}$, let

$$\mathsf{B}_\psi(T) = c^{-1}\int_L \psi_T \wedge *_h\vartheta, \tag{28}$$

where ψ_T is the pull-back of ψ to the link at radius T, regarded as a form on L. Since $*_h\vartheta$ is closed, $\mathsf{B}_\psi(T)$ is the coefficient of ϑ in the harmonic part of ψ_T, and it depends only on the class of ψ_T when ψ_T is closed.

Lemma 4.1. *Let (V_σ, g_V^A) be a cap, with radial function r on its conical part $\{r \ge 2A\}$. There is C_A such that for every $T > 2A$ and every smooth two-form ψ on $V_\sigma\cap\{r\le T\}$,*

$$\int_{\{r\le T\}} \left(|d\psi|^2 + |d^*\psi|^2\right) dV \ge \frac{c\,|\mathsf{B}_\psi(T)|^2}{\log T + C_A}. \tag{29}$$

The left side is unchanged when the metric is multiplied by a constant.

Proof. By (9) there is a closed three-form Ξ on the compact set $\{r \le 2A\}$ whose restriction to the boundary is $*_h\vartheta$. Stokes' theorem and the Cauchy–Schwarz inequality give

$$c\,\mathsf{B}_\psi(2A) = \int_{\{r\le 2A\}} d\psi\wedge\Xi, \qquad \int_{\{r\le 2A\}} |d\psi|^2 \ge \kappa_A |\mathsf{B}_\psi(2A)|^2, \quad \kappa_A = c^2/\|\Xi\|_{L^2}^2.$$

On the conical part write $\psi = \psi_r + dr\wedge\beta(r)$. The component of $d\psi$ along $dr\wedge\vartheta$ is $\mathsf{B}'_\psi(r)\,dr\wedge\vartheta$, because $d_L^*\vartheta = 0$. Since $|dr\wedge\vartheta|^2 = r^{-4}|\vartheta|_h^2$ and the volume density is $r^5 dr\,dV_h$,

$$\int_{\{2A\le r\le T\}} |d\psi|^2 \ge c\int_{2A}^T r\,|\mathsf{B}'_\psi(r)|^2 dr.$$

Applying the Cauchy–Schwarz inequality to $\mathsf{B}_\psi(T) = \mathsf{B}_\psi(2A) + \int_{2A}^T \mathsf{B}'_\psi$ gives

$$|\mathsf{B}_\psi(T)|^2 \le \Big(\kappa_A^{-1} + c^{-1}\log\frac{T}{2A}\Big)\int_{\{r\le T\}} |d\psi|^2,$$

which is (29). In dimension six the integrals of $|d\psi|^2$ and $|d^*\psi|^2$ for a two-form scale by ϵ^{6-6} and ϵ^{6-4-2} under $g\mapsto\epsilon^2 g$. □

4.2. **Central harmonic forms.**

Lemma 4.2. *For the metric g_0^R on M_0:*

1. *every smooth square-integrable closed and coclosed one-form vanishes;*
2. *the smooth square-integrable closed and coclosed two-forms form a space $\mathcal{H}_R$ that maps isomorphically onto V; write η_D^R for the element with class D;*
3. *on the conical part $\{r<R\}$ of the end at p_i,*

$$\eta_D^R = \mathrm{per}_i(D)\,\vartheta_i + d\beta_D, \qquad |\beta_D| = O(r^{\kappa-1}), \quad |d\beta_D| = O(r^{\kappa-2}), \tag{30}$$

   for some $\kappa>0$, with β_D a one-form without dr-component whose restriction to each link section is coclosed, and with differentiated bounds;
4. *these forms lie in the minimal domain of $d+d^*$, that is, they are limits in the graph norm of compactly supported smooth forms.*

Proof. On an exactly conical end write a closed and coclosed two-form as $\psi = a(r) + dr\wedge b(r)$ and decompose $a(r) = c_0(r)\vartheta + d_L v(r)$ with $v(r)$ coclosed on the link; there are no harmonic one-forms on L. Closedness gives $c_0' = 0$ and that $b - v'$ is closed; coclosedness then makes $b-v'$ harmonic, hence zero, and

$$b = v', \qquad r^{-2}\Delta_L v - v'' - 3r^{-1}v' = 0.$$

On the eigenspace of coclosed one-forms on L with eigenvalue $\lambda_L>0$ the solutions are $r^{k_\pm}$ with $k_\pm = -1\pm\sqrt{1+\lambda_L}$. The corresponding two-forms have pointwise size $r^{k_\pm-2}$, and are square integrable near the vertex exactly when $k_\pm > -1$, since their squared radial norm is proportional to $\int_0^{r_0} r^{2k_\pm+1}\,dr$; this excludes $k_- < -2$. This gives (30) with β_D the sum of the terms with $k_+>0$ and $\kappa = \min k_+$; convergence and the differentiated bounds follow from the expansion in eigenforms and elliptic estimates on annuli of fixed ratio. The term $\mathrm{per}_i(D)\vartheta_i$ is identified by its class.

For (4), cut off the term ϑ_i with a function χ_ρ equal to zero for $r\le\rho^2$ and one for $r\ge\rho$, with $|d\chi_\rho|\le C/(r|\log\rho|)$; since ϑ_i is closed and coclosed with $|\vartheta_i| = O(r^{-2})$, the graph-norm error is $O(|\log\rho|^{-1/2})$. Cut off $d\beta_D$ with an ordinary cutoff at scale ρ; the error is $O(\rho^\kappa)$.

For (2), injectivity is Lemma 3.3. For surjectivity consider the closure $d_{\min}$ of d on compactly supported forms. The orthogonal complement in $\ker d_{\min}$ of the closure of the image

of $d_{\min}$ consists of closed and coclosed forms, which are smooth by elliptic regularity. The forms η_D of Proposition 3.2 are closed, square integrable for g_0^R, which is uniformly equivalent to g_0, and lie in the domain of $d_{\min}$ by the cutoff argument just given. Their images in the quotient $\ker d_{\min}/\overline{\operatorname{im} d_{\min}}$ are independent, because the pairings with compactly supported closed four-forms representing a basis of $H^4_c(M_0)$ are continuous in L^2 and vanish on $\overline{\operatorname{im} d_{\min}}$. Hence $\dim \mathcal{H}_R \geq b$.

For (1), the same analysis on an end shows that a closed and coclosed square-integrable one-form is, near the vertex, the differential of a sum of homogeneous harmonic functions with exponents $-2+\sqrt{4+\lambda_L} > 0$, for eigenvalues λ_L of the link; the remaining solutions $r^{-5}dr$ and those with negative exponents are not square integrable. Since $H^1(M_0) = 0$ the form has a global primitive, bounded near the nodes, and the cutoff argument of Lemma 3.3 shows that it vanishes. □

Let $G_R(D,E) = \langle \eta_D^R, \eta_E^R \rangle_{L^2(g_0^R)}$ and let $\mathcal{M}_R$ be its quotient on $\mathcal{Q}$. Each η_D^R minimizes the L^2 norm of g_0^R among closed forms in the domain of $d_{\min}$ with class D, and the same holds for η_D and g_0. Since the two norms differ by a factor $1+O(R)$ by (14),

$$G_R = (1+O(R))\, G, \qquad \mathcal{M}_R = (1+O(R))\, \mathcal{M}, \tag{31}$$

as quadratic forms, and the eigenvalues $\mu_j(R)$ of $\mathcal{E}$ relative to $\mathcal{M}_R$ converge to μ_j as $R \to 0$.

4.3. **The caps.**

Lemma 4.3. *For the complete metric g_V^A on V_σ:*

1. *there are no nonzero square-integrable harmonic forms;*
2. *if ψ is a smooth solution of $(d+d^*)\psi = 0$ with $|\psi| = O(r^{-3})$ at infinity, then the part of ψ that is homogeneous of pointwise order r^{-3} has form degree three.*

Proof. The metric g_V^A is exactly conical outside a compact set. By [HHM04, Theorem 1A] its space of L^2 harmonic forms of degree j is $H^j_c(V_\sigma)$ for $j < 3$, $H^j(V_\sigma)$ for $j > 3$, and the image of $H^3_c(V_\sigma) \to H^3(V_\sigma)$ for $j = 3$. Since $V_\sigma \simeq T^*S^3$, the first two vanish, and the third is zero because the map is the Euler class of the cotangent bundle of S^3. This proves (1).

For (2), first $d\psi = d^*\psi = 0$ separately: for a cutoff χ_T equal to one for $r \leq T$ and supported in $r \leq 2T$, $\|(d+d^*)(\chi_T\psi)\|^2 \leq CT^{-2}\int_{T\leq r\leq 2T} |\psi|^2 \leq CT^{-2}$, and for compactly supported forms this norm is $\|d(\chi_T\psi)\|^2 + \|d^*(\chi_T\psi)\|^2$. A closed and coclosed form of pure degree p, homogeneous of pointwise order r^{-3} on the cone, is $r^{p-3}a + dr \wedge r^{p-4}b$ with a, b on the link and

$$d_L a = 0, \quad (p-3)a = d_L b, \quad d_L^* b = 0, \quad d_L^* a + (p-3)b = 0.$$

For $p \neq 3$ these give $\Delta_L b = -(p-3)^2 b$, so $b = 0$ and $a = 0$. □

4.4. **Compactness.**

Proposition 4.4. *Let $p \in \{1,2\}$ and let $\psi^1_\epsilon, \dots, \psi^n_\epsilon$ be orthonormal eigenforms of degree p on $(\overline{X}_\epsilon, \bar{g}_\epsilon)$ with eigenvalues tending to zero along a sequence $\epsilon \to 0$. Then, after passing to a subsequence, they converge smoothly on compact subsets of M_0, and without loss of mass, in the sense that $\langle \psi^a_\epsilon, \psi^b_\epsilon \rangle \to \langle \psi^a_0, \psi^b_0 \rangle$, to square-integrable closed and coclosed p-forms ψ^a_0 for g_0^R. In particular the limits are orthonormal.*

Proof. The model is of the kind treated by Anné and Takahashi [AT15]: a fixed manifold with exactly conical ends at finitely many points, to which complete manifolds with exactly conical ends, scaled by ϵ, are glued along the cones. Their analysis is carried out for one cap; nothing in it uses connectedness of the cap, and it applies to finitely many, because the

tangential operator, the Poisson extensions and the boundary operators on the caps, and the transmission conditions on the collars [AT15, Sections 2–4] are direct sums over the nodes, and the radial estimates are sums of integrals over disjoint collars. No condition on the spectrum of the link is imposed there. Their decomposition of a family of eigenforms with bounded eigenvalues [AT15, Corollary 17 and Lemma 18] gives, on the central part,

$$\psi_\epsilon^a = \Phi_\epsilon^a + \mathcal{V}_\epsilon^a + \mathcal{C}_\epsilon^a + o_{L^2}(1), \qquad \Phi_\epsilon^a \longrightarrow \psi_0^a \ \text{ in } L^2,$$

where $\mathcal{V}_\epsilon^a$ comes from square-integrable solutions of $(d+d^*)\psi = 0$ on the complete caps and $\mathcal{C}_\epsilon^a$ from solutions with $|\psi| = O(r^{-3})$, normalized by $\ell_\epsilon^{-1/2}$, and the three terms are asymptotically orthogonal:

$$\lim \langle \psi_\epsilon^a, \psi_\epsilon^b \rangle = \langle \psi_0^a, \psi_0^b \rangle + \sum_i \langle v_i^a, v_i^b \rangle + \sum_i \langle \varsigma_i^a, \varsigma_i^b \rangle, \tag{32}$$

with v_i^a the square-integrable cap solutions and ς_i^a the coefficients of the solutions of order r^{-3}. By Lemma 4.3(1) the v_i^a vanish. Since the eigenvalues tend to zero, the limits ψ_0^a are closed and coclosed of degree p. The coefficients ς_i^a are leading coefficients of actual solutions of $(d+d^*)\psi = 0$ on the complete caps with $|\psi| = O(r^{-3})$, by [AT15, Proposition 11] and the first step of the proof of [AT15, Proposition 12]. By Lemma 4.3(2) the leading part of $\mathcal{C}_\epsilon^a$ therefore has degree three; projecting the decomposition onto forms of degree three, where ψ_ϵ^a and ψ_0^a have no component, shows that $\mathcal{C}_\epsilon^a \to 0$ in L^2, so the ς_i^a vanish. Then (32) is the assertion. □

Proposition 4.5. *The models have exactly m positive eigenvalues in degree two tending to zero as $\epsilon \to 0$, and the degree-one spectrum and the remaining positive degree-two spectrum are bounded below by a positive constant depending on R, A and u.*

Proof. By Proposition 4.4 and Lemma 4.2, at most b eigenvalues in degree two and none in degree one tend to zero, counted with the zero eigenvalue. Conversely, by Lemma 4.2(4) an orthonormal basis of $\mathcal{H}_R$ is approximated in the graph norm of $d+d^*$ by compactly supported forms; extended by zero to the model, they show by the min–max principle that at least b eigenvalues tend to zero. The model is diffeomorphic to X_t by Lemma 2.5, so exactly $s = b_2(X_t)$ of them are zero, by Lemma 3.1. □

4.5. **The leading asymptotics.** Let $\bar\lambda_1(\epsilon) \le \cdots \le \bar\lambda_m(\epsilon)$ be the small positive eigenvalues of the model in degree two.

Theorem 4.6. *For fixed R, A and u,*

$$\lim_{\epsilon\to 0} \log(1/\epsilon)\, \bar\lambda_j(\epsilon) = c\, \mu_j(R) \qquad (1 \le j \le m),$$

where $c = \pi$ and $\mu_j(R)$ are the eigenvalues of $\mathcal{E}$ relative to $\mathcal{M}_R$.

Proof. The harmonic space of the model. For $k \in K$ the form η_k^R has no ϑ_i-terms, so by (30) it equals $d\beta_k$ near each node, and $\gamma_{k,\rho} = \eta_k^R - d(\chi_\rho \beta_k)$, with χ_ρ equal to one near the nodes and supported in $\{r < 2\rho\}$, is a compactly supported closed form in the class k with $\|\gamma_{k,\rho} - \eta_k^R\|^2 = O(\rho^{2\kappa+2})$. Extended by zero it is a closed form on the model representing the class in $H^2(\overline{X}_\epsilon)$ that corresponds to k. Hence the harmonic representative $h_\epsilon(k)$ of that class satisfies $\limsup \|h_\epsilon(k)\|^2 \le G_R(k,k)$. Conversely, by Proposition 4.4, applied to the harmonic forms, a subsequence converges without loss of norm to an element of $\mathcal{H}_R$, whose class is k by pairing with compactly supported closed four-forms. Thus the Gram matrix of the harmonic forms of the model on K converges to $G_R|_K$, and the harmonic forms converge in L^2 to the η_k^R.

Upper bound. Let D be G_R-orthogonal to K. With

$$\chi_\epsilon(r) = \frac{\log(r/2A\epsilon)}{\log(R/2A\epsilon)} \quad (2A\epsilon \le r \le R),$$

extended by zero and one, put $T_\epsilon D = \chi_\epsilon \eta_D^R$, extended by zero over the caps; it lies in the form domain of the model. The metric is exactly conical where χ_ϵ varies, and $dT_\epsilon D = d\chi_\epsilon \wedge \eta_D^R$, $d^* T_\epsilon D = -\iota_{\nabla\chi_\epsilon}\eta_D^R$. By (30) and $\iota_{\partial_r}\vartheta_i = 0$, the leading term is $d\chi_\epsilon \wedge \mathrm{per}_i(D)\vartheta_i$, of squared norm $c\,\mathrm{per}_i(D)^2 \int r|\chi_\epsilon'|^2 dr = c\,\mathrm{per}_i(D)^2/\log(R/2A\epsilon)$; the cross term with $d\chi_\epsilon \wedge d\beta_D$ vanishes after integration over the link because the tangential part of $d\beta_D$ is exact on the link and ϑ_i is coclosed; and the remaining terms are $O(\ell_\epsilon^{-2})\int_0^R r^{2\kappa-1}dr$. Write $\bar q_\epsilon(\psi) = \|d\psi\|^2 + \|d^*\psi\|^2$ for the energy of a form on the model. Then

$$\begin{aligned} \bar q_\epsilon(T_\epsilon D) &= \frac{c\,\mathcal{E}([D],[D])}{\ell_\epsilon} + O(\ell_\epsilon^{-2})\|D\|^2, \\ \langle T_\epsilon D, T_\epsilon E\rangle &= \mathcal{M}_R([D],[E]) + O(\ell_\epsilon^{-1})\|D\|\|E\|; \end{aligned} \tag{33}$$

the second holds because $|1-\chi_\epsilon| \le C(1+|\log r|)/\ell_\epsilon$ is integrable against $|\eta_D^R|^2 dV = O(r\,dr)$, and because $G_R = \mathcal{M}_R$ on the G_R-orthogonal complement of K. Subtract from $T_\epsilon D$ its projection onto the harmonic forms of the model. By the first step and $G_R(D,K) = 0$ this changes the norm by $o(1)$, and it does not change $\bar q_\epsilon$. The min–max principle on the span of the first j eigenvectors of $(\mathcal{E}, \mathcal{M}_R)$ gives

$$\limsup_{\epsilon\to 0} \ell_\epsilon \bar\lambda_j(\epsilon) \le c\,\mu_j(R).$$

Lower bound. Choose a sequence along which $\ell_\epsilon\bar\lambda_j(\epsilon)$ tends to its lower limit ℓ. By Proposition 4.4 the first j orthonormal eigenforms $\psi_\epsilon^1, \dots, \psi_\epsilon^j$ have, along a subsequence, orthonormal limits $\psi_0^a \in \mathcal{H}_R$, which are L^2-orthogonal to the η_k^R, $k \in K$, by the first step. For $z \in \mathbb{R}^j$ put $\psi_\epsilon(z) = \sum z_a \psi_\epsilon^a$ and $\psi_0(z) = \sum z_a \psi_0^a = \eta_{D(z)}^R$, with $D(z)$ in the G_R-orthogonal complement of K and $\mathcal{M}_R([D(z)]) = |z|^2$. At the fixed sections $\{r = R/2\}$ the forms converge smoothly, so $\mathsf{B}_{\psi_\epsilon(z)}(R/2) \to \mathrm{per}_i(D(z))$ at the node p_i. The part of the model inside this section at p_i is isometric to a truncated cap with the metric $\epsilon^2 g_V^A$, by an identification that differs from the chart by the unit u and so preserves ϑ_i, and Lemma 4.1 gives

$$\bar\lambda_j(\epsilon)|z|^2 \ge \bar q_\epsilon(\psi_\epsilon(z)) \ge \sum_i \frac{c\,|\mathsf{B}_{\psi_\epsilon(z)}(R/2)|^2}{\log(R/2\epsilon) + C_A}.$$

Multiplying by ℓ_ϵ and passing to the limit, $c\,\mathcal{E}([D(z)]) \le \ell\,\mathcal{M}_R([D(z)])$ for all z. The classes $[D(z)]$ fill a j-dimensional subspace of $\mathcal{Q}$, so $\ell \ge c\,\mu_j(R)$. □

5. Proof of the main theorem

Lemma 5.1 (Dodziuk)**.** *Let $g, \bar g$ be smooth metrics on a closed n-manifold with $(1-\delta)\bar g \le g \le (1+\delta)\bar g$, $0 < \delta < 1/2$. Then the ordered positive eigenvalues of the Hodge Laplacians in each degree p satisfy*

$$\Theta(\delta)^{-1}\lambda_{p,j}^+(\bar g) \le \lambda_{p,j}^+(g) \le \Theta(\delta)\,\lambda_{p,j}^+(\bar g), \qquad \Theta(\delta) = 1 + O_n(\delta).$$

Proof. This follows from [Dod82, Proposition 3.3]. The positive spectrum in degree p is the union of the squared singular values of d on $\mathrm{Dom}\,d/\ker d$ in degrees p and $p-1$, with the quotient L^2 norms; these norms and the norms of the images are changed by factors $1+O_n(\delta)$, and neither the operator d nor its domain depends on the metric. □

Proof of Theorem 1.1. Let $t_l \to 0$ be a sequence, and write $t_l = \epsilon_l^2 u_l^2$ as in (11). Passing to a subsequence, $u_l \to u_0$. Fix R and A. By Lemma 2.5, X_{t_l} is identified with $\overline{X}_{\epsilon_l,u_0}$ in such a way that g_{t_l} and $\bar{g}^{R,A}_{\epsilon_l,u_0}$ differ, relative to the latter, by at most δ_l with $\limsup_l \delta_l \le \delta(R,A) := C(R + A^{-2})$. Both manifolds have s zero eigenvalues in degree two and none in degree one. By Lemma 5.1, Proposition 4.5 and Theorem 4.6, if $\delta(R,A) < 1/4$ then for large l exactly m positive degree-two eigenvalues of g_{t_l} are small, the degree-one spectrum and the other positive degree-two eigenvalues are bounded below by a positive constant, and

$$\Theta(2\delta(R,A))^{-1} c\,\mu_j(R) \le \liminf_l \log(1/\epsilon_l)\lambda^+_{2,j}(t_l)$$
$$\le \limsup_l \log(1/\epsilon_l)\lambda^+_{2,j}(t_l) \le \Theta(2\delta(R,A))\, c\,\mu_j(R).$$

Let $R \to 0$ and $A \to \infty$ and use (31): along the subsequence, $\log(1/\epsilon_l)\lambda^+_{2,j}(t_l) \to c\,\mu_j = \pi\mu_j$. Since $\log(1/\epsilon) = \frac{1}{2}\log(1/|t|)$ this is (3) along the subsequence. As every sequence has such a subsequence, (3) holds, and so do the count and the uniform lower bound for the remaining spectrum.

Let ψ be an eigenform for a small eigenvalue λ, with $|t|$ so small that λ is below the bound γ for the degree-one spectrum. Then $d^*\psi$ is a one-form with $\Delta d^*\psi = \lambda d^*\psi$, hence zero, and $\psi = \lambda^{-1} d^* d\psi$ is coexact. Since g_t is Kähler, Λ commutes with Δ, so $\Lambda\psi$ is a function with $\Delta\Lambda\psi = \lambda\Lambda\psi$. The positive spectrum on functions is the spectrum on exact one-forms and is at least γ; hence $\Lambda\psi = 0$. The type is the subject of the next proposition. □

Proposition 5.2. *For $|t|$ sufficiently small, every eigenform of (X_t, g_t) for one of the eigenvalues $\lambda^+_{2,1}(t), \dots, \lambda^+_{2,m}(t)$ is of type $(1,1)$.*

Proof. Let S_t be the span of these eigenforms and P_t the pointwise orthogonal projection of two-forms on X_t onto the forms of type $(2,0)+(0,2)$. Since g_t is Kähler, P_t commutes with Δ; it therefore preserves S_t, and its restriction to S_t is an orthogonal projection. It suffices to show that $\|P_t\psi\| \le \frac{1}{2}$ for all unit $\psi \in S_t$ and all small t. Suppose not; then there are $t_l \to 0$ and unit $\psi_l \in S_{t_l}$ with $\|P_{t_l}\psi_l\| > \frac{1}{2}$. Fix R, A and pass to a subsequence as in the preceding proof, so that $g = g_{t_l}$ is a metric on $\overline{X}_{\epsilon_l,u_0}$ within relative distance $\delta \le 2\delta(R,A)$ of the model metric $\bar{g}$. Constants C below are independent of R, A and l.

Comparison with the small eigenforms of the model. Let $\mathcal{Z}$ be the space of closed square-integrable two-forms and $\bar{\psi}_l$ the component of ψ_l orthogonal to $\mathcal{Z}$ for $\bar{g}$. Then $d\bar{\psi}_l = d\psi_l$ and $\bar{\psi}_l$ is coexact for $\bar{g}$. Since ψ_l is orthogonal to $\mathcal{Z}$ for g,

$$\|\bar{\psi}_l\|_{\bar{g}} = \operatorname{dist}_{\bar{g}}(\psi_l, \mathcal{Z}) \ge (1 - C\delta)\operatorname{dist}_g(\psi_l, \mathcal{Z}) = 1 - C\delta, \qquad \|\psi_l\|_{\bar{g}} \le 1 + C\delta,$$

so $\|\psi_l - \bar{\psi}_l\|^2_{\bar{g}} = \|\psi_l\|^2_{\bar{g}} - \|\bar{\psi}_l\|^2_{\bar{g}} \le C\delta$. Moreover $\|d\bar{\psi}_l\|^2_{\bar{g}} \le (1 + C\delta)\lambda^+_{2,m}(t_l) \to 0$. By Proposition 4.5 the coexact two-forms of the model have m small eigenvalues and a gap above them, so the distance in $L^2(\bar{g})$ from $\bar{\psi}_l$ to the span $\bar{S}_l$ of the corresponding eigenforms tends to zero.

Limits. By Proposition 4.4 and the first step of the proof of Theorem 4.6, after passing to a subsequence, unit elements of $\bar{S}_l$ converge, smoothly on compact subsets of M_0 and without loss of L^2 norm, to forms η^R_D with D orthogonal to K for G_R. For such D the form $\eta^R_D - \eta_D$ lies in the closure of the exact forms in the domain of $d_{\min}$, to which η_D is orthogonal for g_0, and η^R_D minimizes the norm of g^R_0 in its class; with (14) this gives

$$\|\eta^R_D - \eta_D\|^2_{g_0} = \|\eta^R_D\|^2_{g_0} - \|\eta_D\|^2_{g_0} \le CR\,\|\eta_D\|^2_{g_0}.$$

The forms η_D are of type $(1,1)$ on M_0 by Proposition 3.2.

Conclusion. For $\rho > 0$ let $W_\rho = M_0 \cap \{r \geq \rho\}$. The identifications Φ_t of Lemma 2.5 restrict to embeddings of W_ρ into X_t that depend smoothly on t and are the inclusion at $t = 0$, so the complex structures $\Phi_t^* J_t$ converge smoothly to that of M_0 on W_ρ, and so do the projections P_t to the projection P_0 on M_0. Combining the two steps, on W_ρ the forms ψ_l are within $C(\delta^{1/2} + R^{1/2}) + o(1)$ in L^2 of a form η_D of norm at most 2, for which $P_0 \eta_D = 0$; and the L^2 norm of ψ_l on the complement of W_ρ is at most $C(\delta^{1/2} + R^{1/2}) + o(1) + \sup \|\eta_D\|_{L^2(\{r<\rho\})}$, the supremum over D with $G(D, D) \leq 4$, which tends to zero with ρ. Since $|P_t \psi| \leq |\psi|$ pointwise, choosing first R small and A large, then ρ small, gives $\|P_{t_l} \psi_l\| < \frac{1}{2}$ for large l, a contradiction. □

6. Examples and remarks

Example 6.1 (Determinantal hypersurfaces)**.** Let $d \geq 2$, let l_1, l_2 be independent linear forms on $\mathbb{P}^4$ and f_1, f_2 general forms of degree $d - 1$, and put

$$X_0 = \{F_0 = 0\}, \quad F_0 = l_1 f_2 - l_2 f_1, \qquad \mathcal{X} = \{(t, x) \in \mathbb{D} \times \mathbb{P}^4 : \ F_0(x) + tP(x) = 0\},$$

where P is a form of degree d that does not vanish at the points of $\{l_1 = l_2 = f_1 = f_2 = 0\}$, and $\mathbb{D}$ is a small disc. Let ω be the restriction to $\mathcal{X}$ of $\frac{i}{2} dt \wedge d\bar{t} + \omega_{FS}$, where ω_{FS} is the Fubini–Study form with $[\omega_{FS}] = c_1(\mathcal{O}(1))$; thus g_t is the Fubini–Study metric on the hypersurface X_t.

The singular set of X_0 consists of the $(d-1)^2$ points of $\{f_1 = f_2 = 0\}$ in the plane $\{l_1 = l_2 = 0\}$, which are ordinary double points: there (l_1, l_2, f_1, f_2) are local coordinates on $\mathbb{P}^4$. The total space $\mathcal{X}$ is smooth at these points because $\partial_t(F_0 + tP) = P \neq 0$, and smooth elsewhere near X_0. A small resolution is the complete intersection

$$Y = \{\zeta_0 l_1 + \zeta_1 l_2 = 0, \ \zeta_0 f_1 + \zeta_1 f_2 = 0\} \subset \mathbb{P}^4 \times \mathbb{P}^1,$$

of the ample classes $H + L$ and $(d-1)H + L$, where H and L are the hyperplane classes of the factors. The first divisor is the blow-up of $\mathbb{P}^4$ along the plane, on which the second linear system is base-point free, so Y is smooth for general f_1, f_2 by Bertini's theorem, and X_0 is smooth away from the $(d-1)^2$ points because $Y \to X_0$ is an isomorphism there. By the Lefschetz theorem, applied twice, $b_1(Y) = 0$ and $H^2(Y; \mathbb{Z}) = \mathbb{Z}H \oplus \mathbb{Z}L$, and likewise $b_2(X_t) = 1$. Thus the hypotheses of Theorem 1.1 hold, with $K = \mathbb{R}H$ and $m = 1$. From the class of Y,

$$\int_Y H^3 = d, \qquad \int_Y H^2 L = d - 1, \qquad \int_Y H L^2 = 0,$$

and each exceptional curve is a fibre of the projection to $\mathbb{P}^4$, so $H \cdot C_i = 0$ and $L \cdot C_i = 1$. By (27) with $D = L$,

$$\mu_1 = \frac{(d-1)^2}{(d-1)^2/d - 0} = d, \qquad \lambda^+_{2,1}(t) = \frac{2\pi d}{\log(1/|t|)} \, (1 + o(1)).$$

For $d = 5$ this is the split quintic, and the coefficient 10π is the one found in [W26] for the Ricci-flat metrics in the class $c_1(\mathcal{O}(1))$.

Remark 6.2 (Dependence on the metric)*.* The limits in Theorem 1.1 depend on ω only through the class H. For the Ricci-flat metrics of a polarized Calabi–Yau conifold transition the same limits are proved in [W26], with the logarithm taken along a ray. There the tangent cone at a node is the Ricci-flat conifold, with radial function homogeneous of degree $2/3$ in the coordinates of (5); the constant (8) is $2\pi/3$ and the scale is $|t|^{1/3}$, in place of π and $|t|^{1/2}$, and the product $c/\log(1/\epsilon) = 2\pi/\log(1/|t|)$ is the same. In both cases $\|\vartheta\|^2_{L^2(L)} =$

$2\pi\alpha \cdot (-\int_{Z_\sigma} \vartheta_Z^2) = \pi\alpha$ and $\epsilon = |t|^{\alpha/2}$, where α is the degree of homogeneity of the radial function in the coordinates of (5) ($\alpha = 1$ here and $\alpha = 2/3$ for the Ricci-flat cone).

Remark 6.3 (The form G). The form G of (21) agrees with the Hodge–Riemann form of the class H on the primitive classes and is positive on $\mathbb{R}H$: it is $-\int_Y D^2H$ on classes with $\int_Y DH^2 = 0$, and $\frac{1}{2}\int_Y H^3$ on H. The resolution Y enters only through its cohomology ring; the form B is unchanged by flops of the curves C_i, since $H \cdot C_i = 0$, and per changes by signs, so $\mathcal{E}$, $\mathcal{M}$ and the μ_j are unchanged.

Remark 6.4 (Scope). The theorem concerns one-parameter families with smooth total space, so that every node is smoothed to first order; families in which the nodes are smoothed at different rates, and the next term in (3), are not addressed. The small two-form eigenvalues are accompanied by eigenvalues of the same size on exact three-forms and on their Hodge duals, and we do not determine the complete small spectrum in degree three.

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